

EXPERIMENT 6b – DATA VISUALIZATION USING POWER BI

Aim

To load the tourism dataset into Power BI, transform the data, and visualize it to obtain useful insights about tourist locations, visitors, ratings, and revenue.

ALGORITHM 1: IMPORTING DATA INTO POWER BI

Step 1

Open **Power BI Desktop**.

Step 2

Select:

Home → Get Data

Step 3

Choose:

Text/CSV

Step 4

Browse and select the dataset:

tourism_dataset (1).csv

Step 5

Click **Load**.

Step 6

Verify the imported data and its columns:

- Location
- Country
- Category
- Visitors
- Rating
- Revenue
- Accommodation_Available

ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY

Step 1

Select **Transform Data**.

Step 2

Open **Power Query Editor**.

Step 3

Remove duplicate records if any are present.

Step 4

Handle null or blank values in the dataset.

Step 5

Check and set the correct data types:

- Location → Text
- Country → Text
- Category → Text

- Visitors → Whole Number
- Rating → Decimal Number
- Revenue → Decimal Number
- Accommodation_Available → Text

Step 6

Apply the changes and click:

Close & Apply

ALGORITHM 3: DATA FILTERING AND PREPARATION

Step 1

Check the **Visitors** column for valid numerical values.

Step 2

Check the **Rating** column and ensure it contains numerical values.

Step 3

Check the **Revenue** column and ensure the correct data type is applied.

Step 4

Standardize text values in columns such as **Country** and **Category**.

Step 5

Remove unnecessary blank or invalid records if present.

Step 6

Click **Close & Apply** to load the cleaned dataset into the Power BI model.

ALGORITHM 4: CREATING CALCULATIONS

Step 1

Navigate to **Report View** or **Table View**.

Step 2

Select the tourism dataset table.

Step 3

Create a measure for **Total Visitors**:

Total Visitors = SUM('tourism_dataset'[Visitors])

Step 4

Create a measure for **Average Rating**:

Average Rating = AVERAGE('tourism_dataset'[Rating])

Step 5

Create a measure for **Total Revenue**:

Total Revenue = SUM('tourism_dataset'[Revenue])

Step 6

Press **Enter** to validate and calculate the measures.

ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING

Step 1

Select a **Bar Chart** and map:

- Location → Axis
- Visitors → Values

Step 2

Select a **Donut Chart** and map:

- Category → Legend
- Category → Values

Step 3

Add **Card** visuals for:

- Total Visitors
- Average Rating
- Total Revenue

Step 4

Add **Slicer** visuals for:

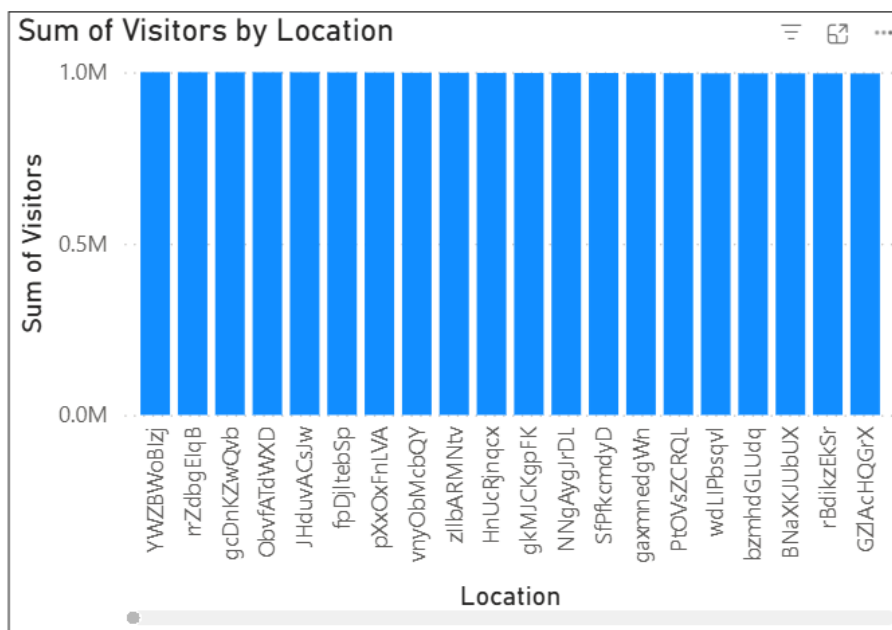
- Country
- Category

to enable interactive filtering.

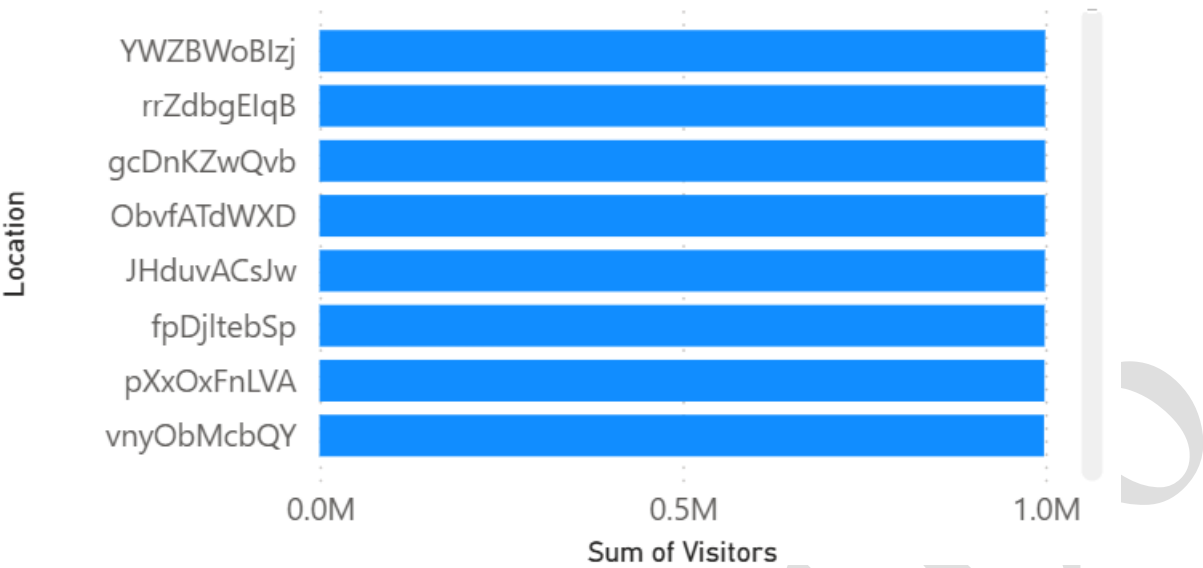
Step 5

Arrange all the visuals to create the Tourism Dashboard.

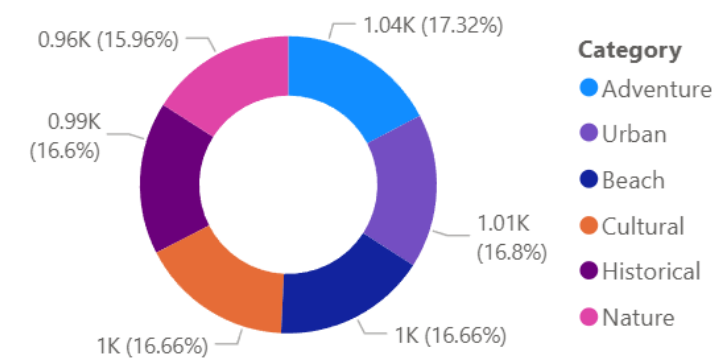
OUTPUT:



Sum of Visitors by Location



Count of Category by Category



RESULT: